

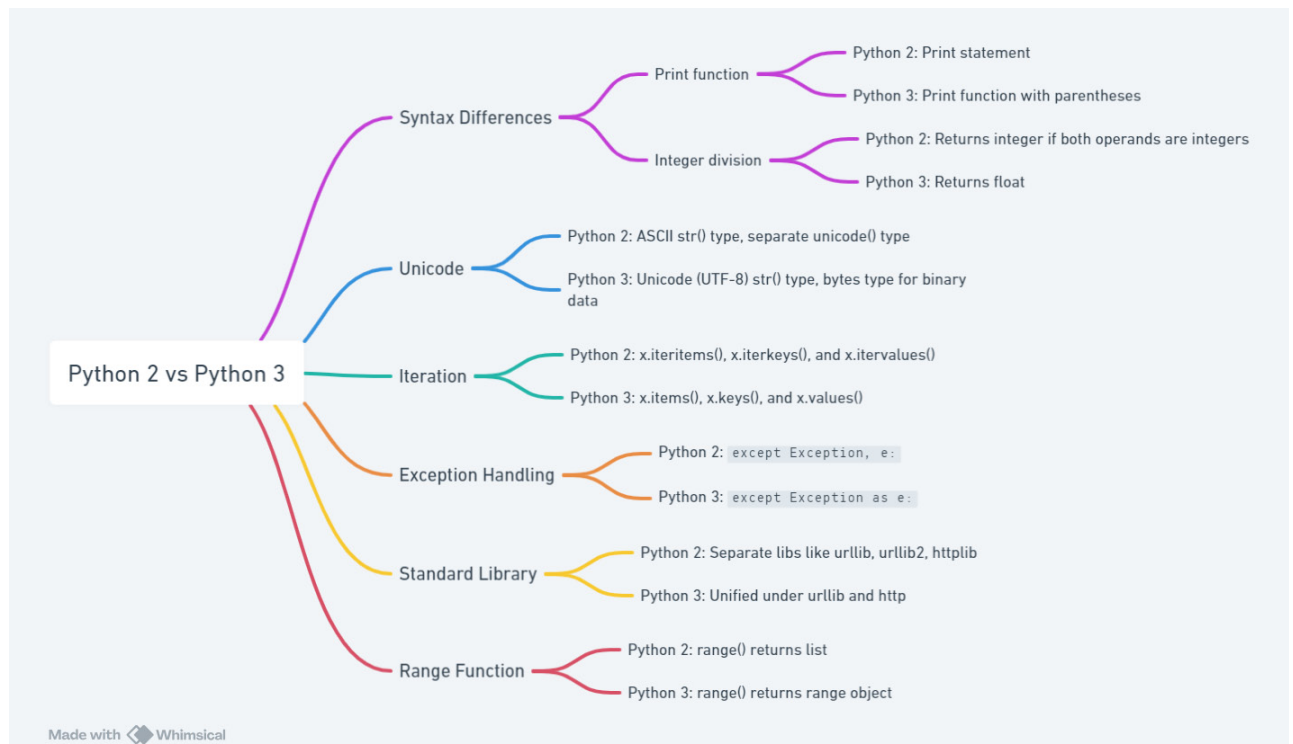


Figure 2: Python 2 vs Python 3

Python 3: In Python 3, integer division returns a float if necessary, so $7/3$ would result in `2.3333333333333335`.

```
division_result = 7/3
print(division_result)
#Outputs: 2.3333333333333335
```

Error handling: Python 3 now requires the `'as'` keyword when you implement error handling. This change separates error handling syntax from function parameter syntax and makes error handling code easiest to read at a glance. Many developers transitioning from Python 2 to Python 3 overlook this small but important change.

Python 2:

```
try:
    trying_to_check_error
except NameError, err:
    print err, <Error Caused>
```

Python 3:

```
try:
    trying_to_check_error
except NameError as err: # new
    'as' keyword
    print (err, <Error Caused>)
```

Security improvements: Python 3 incorporates various security enhancements, including better hash algorithms and protections against common security vulnerabilities. By prioritising security, it offers a safer development environment, reducing the risk of potential exploits and vulnerabilities in deployed applications.

Performance optimisations: Python 3 introduces performance optimisations at both the interpreter and standard library levels, resulting in improved runtime performance and reduced resource consumption. These optimisations

enhance the overall efficiency and scalability of Python applications, making Python 3 a compelling choice for high-performance computing tasks and large-scale deployments.

Python 2 vs Python 3

Choosing between Python 2 and 3 can be challenging, especially if you're just starting out. However, there are some things to keep in mind that might help you decide.

If you're starting a new project from scratch, you should use Python 3. It's the future of the language, and it offers some benefits over Python 2 that can make your life easier in the long run.

However, if you need to maintain compatibility with Python 2 code or libraries, you'll need to use Python 2 for your project. In this case, it's probably best to learn both versions to be as versatile as possible. 

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